CMOS Analog integrated circuit design

-Lab 3- Final Project-Two Stage Opamp

Design Assignment

You will need to design a two stage differential-to-single-ended opamp, either a folded-cascode topology or a two-stage architecture with the following specifications. This design must be implemented in HLMC 55nm process in Server MOSFET.

Specification

The following specification has to be met, over the temperature from -40°C to +105°C, the process ff/tt/ss corner and the supply voltage from 1.1V to 1.3V.

Spec	Value	Unit
VDD	1.2	V
A_0	≥ 60	dB
fu	≥ 40	MHz
SR	≥ 10	V/μs
CL	10	pF
Output Swing	≥ 0.6	V
PM	≥ 60	°

Extra design bonus will be given:

Total Current (IDD): as small as possible

FOM: Figure of Merit $(FOM) = GBW \times C_L/IDD(MHz \times pF/mA)$ for your opamp. The higher the FOM, the higher score you may get.

Specs must be met over the temperature from -40°C to +105°C, the process ff/tt/ss corner and the supply voltage from 1.1V to 1.3V.

Auxiliary circuits

To bias the circuitry, the current source designed in lab2 can be used as the reference current IREF. All other biasing currents must be developed based on I_{REF} using current mirrors.

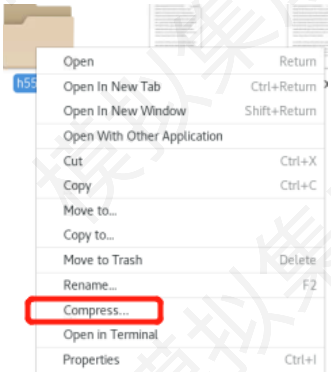
Deliverables:

Design file

All design files and cellviews have to be in a single library and folder, then to be tar+zipped with the following command and file format

tar -czf YourAccountName.tar.gz PathOfYourDesignLibrary, for instance, tar -czf hansy22.tar.gz ~/CMOSICDESIGN2023

Another way to compress files:



Design report:

A maximum 4 pages report has to be submitted via CANVAS system, with the following key sections:

i. Title, Abstract, Index, Author

ii. Introduction: why this work is worth designing;

iii. Architecture: how to determine the topology

iv. Design approach

v. Design calculations

vi. Complete circuit schematic with device dimensions, passive component values and test bench with simulation conditions.

vii. Simulation results (better if including estimated parasitics): tables and plots

viii. Discussions/conclusions: what you have done, what yet have to be done;

ix. References.

Basic Design guideline

Stability and Slew rate oriented design procedure, for a two stage miller compensated opamp, Ref CMOS Analog Circuit Design, 3rd edition, By P Allen.

VDD=1.2V

$A_0 \geq 60$dB

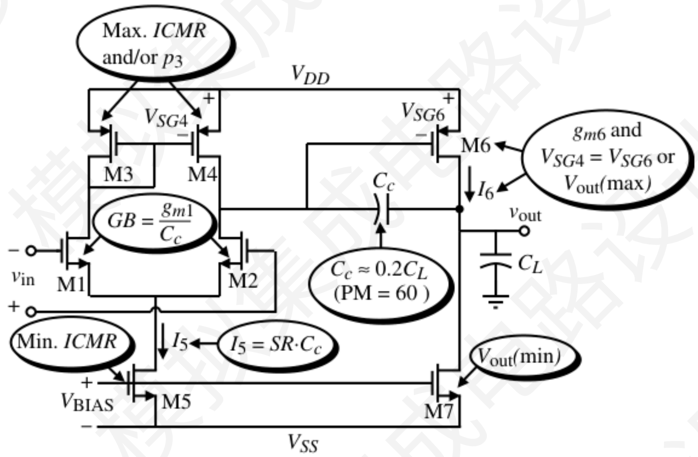
fu /GBW ≥ 40MHz

SR ≥ 10 V/μs

CL = 10pF

Output Swing ≥ 0.6V

PM≥ 60°



1. Choose the smallest device length that will keep the channel modulation parameter constant and give good matching for current mirrors. For devices requries critical matching, please start with the W or L at least 3 times than the minimum W or L. For instance, the input differential pair, current mirrors. In pratical designs, the input device should be kept into minimum, as long as the PVT and MC simulation verify the performance spread.
2. Determine the Differential pair and output common source amplifier.

(1) From the desired phase margin, choose the minimum value for C_C; that is, for a 60° phase margin we use the following relationship.

$$C_c > 0.22C_L$$

(2) Determine the minimum value for the "tail current" (I5) from the largest of the two values.

(3) Determine the g_{m1} from GBW

(4) From 3 and 4, estimate the W/L for input pairs. (In deep submicron process, you may need to look up the simulation curve, rather than the simple square law)

(5) Check if g_m/I_d is ok * (Large enough $>1_5$ for small overdrive and noise), if not increase the g_m

(6) Guarantee the stability, make sure p_2 move beyond 2.2 times of GB.

$$g_{m6} = 2.2 g_{m2}(C_L/C_c)$$

(7) Determine the minimum value for the “tail current” (I_6) from the largest of the two values, for slew rate requirement.

(8) From 6 and 7, calculate the W/L for 6. (In deep submicron process, you may need to look up the simulation curve, rather than the simple square law)

(9) Check if g_m/I_d is ok * (Large enough $>1_5$ for small overdrive and noise),

(10) Check if the gain is ok, if not, increase the g_m.

3. Design the current mirrors

(11) Calculate the required V_{gs6}, and check if V_{gs3} is equal to V_{gs6}

(1) Push the mirror pole and zero due to C_{gs3} and C_{gs4} to be larger than 10 times of GBW. Note that a zero exists at $2p_3$ and will diminish the influence of p_3 on the op amp.

(2) Also make sure that the W/L of M3/M4 and M5/M7 will be sufficiently large to minimize the overdrive voltage, such that the input and output have large swing.

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